

Professional Summary

I am a Principal Software Engineer with 15+ years of experience building cloud-native, distributed systems at scale. I bring deep expertise in Java, Python, Go, Spring Boot, Flask, AWS, Kubernetes, and Terraform, with a proven track record leading complex microservice architectures, cloud migrations, and platform engineering initiatives. I take end-to-end ownership of high-throughput, fault-tolerant systems and full-stack applications across enterprise environments.

Technical Skills

- Languages: Java, Python, Golang, JavaScript, Bash, ZSH
- AI Experience: Claude Code, Gemini, GitHub Copilot, Agentic Workflows using Google Antigravity, Claude Cowork, OpenClaw.
- Cloud: AWS (Lambda, IAM, ECS, EC2, RDS, API Gateway, DynamoDB, CloudWatch, EventBridge, Glue, Step Functions), OpenShift, Cloudfoundry,
- Event Driven Tools: RabbitMQ, AWS SNS, AWS SQS, Webhooks, SSE, WebSocket
- DevOps: Docker, Linux, GitHub, GitLab CI, Jenkins, Terraform, Ansible, UCD, Kubernetes, OpenShift, Vagrant, Docker Compose, Podman, Portainer
- Databases: MySQL, PostgreSQL, Snowflake, DynamoDB, Couchbase, Oracle, Redis, HBASE
- Frameworks/Tools: Spring Boot, Flask, FastAPI, NodeJS, GraphQL, VueJS, RabbitMQ, Apache Spark, Airflow, Tailwind CSS, Bootstrap, ServiceNow
- Testing: Unit testing, Integration testing, Smoke testing, Load testing, Behavior Driven Testing, Gherkin, TestNG, Pytest, Mockito, SoapUI
- Observability: Splunk, Flame Graph, New Relic, Nagios, AppDynamics, CloudWatch, SonarQube
- Misc: Microservices, Distributed Systems, Parallel Systems, OOP, API Design, API Security, Vulnerability analysis, Code Reviews, Client Communication

Work Experience

Capital One (Wilmington, DE) - Senior Software Developer (Data Engineering) Jan 2025 - Present

- Designed, developed, and deployed Java, Spring Boot apps to create and manage large-scale data pipelines of ETL jobs using Spark, AWS Glue, and RDS.
- Wrote unit test cases for ETL jobs with Mockito and BDD test cases with Gherkin, making sure the functionality is intact every iteration.
- Created CI/CD pipelines with GitHub and provisioned cloud infrastructure to automate application build, test, and deployment workflows.
- Created a productivity tool with VueJS & Python to monitor and minimize manual efforts during application troubleshooting from 2 hours to 10 minutes.
- Used Claude Code and GitHub Copilot to build features, write tests and troubleshoot development issues. Used agentic skills with LLM agents via md files.

T. Rowe Price (Owings Mills, MA) - Principal Software Engineer (Global Product Trust) Sep 2023 - Dec 2024

- Developed and deployed Java, Spring Boot, and Python services to AWS ECS utilizing AWS RDS and Lambda using GitLab CI/CD and Terraform.
- Developed sidecar container in Golang with Hashi Corp Vault to fetch credentials and make them available to microservices and lambda.
- Implemented ServiceNow business rules, integrated with Java APIs, and wrote DevOps utilities. Monitored systems using New Relic and Flame Graphs.
- Integrated enterprise change data log with Snowflake to keep a history of transactions and monitor the process by means of dashboards and scripts.
- Improved financial data processing response time from 5 hours to 90 seconds using parallelization, query optimization, and lazy loading.

T-Mobile (Atlanta, GA) - Senior Software Engineer (Cloud Security Domain) Nov 2021 - Sep 2023

- Designed security patterns for APIs complying with T-Mobile standards and OWASP. Developed microservices using Golang for API grading for the same.
- Designed a data harvesting solution to intercept real-time Apigee transaction data for API security grading, utilizing microservices in Java, RabbitMQ, AWS DynamoDB, NodeJS, and AWS Lambda. Interacted with multiple stakeholders to achieve the common goal utilizing traffic in the new solution.
- Worked with DevOps pipelines with GitLab CI/CD. Written Terraform scripts to deploy infra and services. Managed on-prem Kubernetes clusters.
- Modified the groovy codebases utilized as IAC to use upgrade the SDKs making it compatible with new language standards and features.

Fannie Mae (Reston, VA) - Principal Software Engineer (Cloud Migration & Platform Engineering) May 2021 - Nov 2021

- Led a 10-person team for migration efforts from Autosys to AWS(Rehosting), designing solutions with Python and AWS services.
- Built automation pipelines using Jenkins and UCD. Created scalable solutions leveraging AWS Lambda, Step Functions, and ECS.
- Designed and deployed cloud infrastructure for Airflow and Java-based job orchestration in AWS, ensuring end-to-end visibility of the customer journey.
- Solved the SSO dilemma with AWS-hosted Airflow and Fannie Mae SSO, onboarding consumers migrating from AutoSys to Airflow.
- Built a webapp using VueJS and Python in 2 weeks' time to sunset the old application that used to manage network devices fixing a lot of vulnerabilities.

Comcast (Philadelphia, PA) - Senior Fullstack Engineer (Video Entitlement & Site Reliability) Jun 2014 - May 2021

- Led development of a video entitlement highly distributed microservice using Java and Spring Boot, managing e2e deployment, code quality, and reviews.
- Designed and implemented DataGrid, an event-driven caching solution using Java and Couchbase database, reducing billing system call frequency.
- Built Genesis and GoldenEye, data orchestration solutions in Golang and GraphQL Federation, to proactively address customer issues.
- Developed the Spider UI in VueJS and Golang, improved system monitoring with Nagios and AppDynamics.

COMPUNNEL (NJ) - Principal Software Engineer May 2021 - Present

- Working full-time as a Principal Software Engineer for different clients from May 2021 to present. The current client is Capital One.

TCS (India) - Assistant Consultant Dec 2010 - May 2021

- Worked as an Assistant Consultant in this position in India and worked with several clients like SBI and TDC. Worked at Comcast from 2014 - 2021.
- Led the migration of the KVIKOC platform from JSF to AngularJS and executed a critical WebLogic server upgrade to version 10.3.2, resolving performance bottlenecks and enhancing stability.
- Developed cross-border web services using the JOLT protocol to interface with Amdocs billing systems across three countries.
- Designed a custom WebLogic monitoring portal to improve system oversight and deployed real-time financial reporting tools to deliver dynamic contents.

Projects

<https://github.com/susamn/helpful-tools-v2> - Open source, privacy-focused productivity tools for developers, including JSON, regex, cron utilities built using modern web stack, actively maintained and used.

<https://github.com/susamn/shreddem> - Mass-delete unwanted emails in seconds. Shreddem caches your Gmail headers locally so you can instantly filter by sender, find unread bloat, and clean up thousands of emails with one click.

Certifications

Building Generative AI-Powered Applications with Python • Generative AI with Large Language Models • Introduction to Generative AI • AWS Certified Developer - Associate • Sun Certified Java 6 Developer

Education

Bachelor of Instrumentation & Electronics Engineering, Jadavpur University, Kolkata, India (2005 - 2009)